UIL Computer Science Written Test

2026 Invitational A

January 2026

General Directions (Please read carefully!)

1. DO NOT OPEN THE EXAM UNTIL TOLD TO DO SO.
2. There are 40 questions on this contest exam. You will have 45 minutes to complete this contest.
3. All answers must be legibly written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Clean erasures are necessary for accurate grading.
4. You may write on the test packet or any additional scratch paper provided by the contest director, but NOT on the answer sheet, which is reserved for answers only.
5. All questions have ONE and only ONE correct answer. There is a 2-point penalty for all incorrect answers.
6. Tests may not be turned in until 45 minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your test until told to do otherwise. You may use this time to check your answers.
7. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
8. All provided code segments are intended to be syntactically correct, unless otherwise stated. You may also assume that any undefined variables are defined as used.
9. A reference to many commonly used Java classes is provided with the test, and you may use this reference sheet during the contest. AFTER THE CONTEST BEGINS, you may detach the reference sheet from the test booklet if you wish.
10. Assume that any necessary import statements for standard Java SE packages and classes (e.g., `java.util`, `System`, etc.) are included in any programs or code segments that refer to methods from these classes and packages.
11. NO CALCULATORS of any kind may be used during this contest.

Scoring

1. Correct answers will receive **6 points**.
2. Incorrect answers will lose **2 points**.
3. Unanswered questions will neither receive nor lose any points.
4. In the event of a tie, the student with the highest percentage of attempted questions correct shall win the tie.

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

package java.lang

```
class Object
  boolean equals(Object anotherObject)
  String toString()
  int hashCode()

interface Comparable<T>
  int compareTo(T anotherObject)
    Returns a value < 0 if this is less than anotherObject.
    Returns a value = 0 if this is equal to anotherObject.
    Returns a value > 0 if this is greater than anotherObject.

class Integer implements Comparable<Integer>
  Integer(int value)
  int intValue()
  boolean equals(Object anotherObject)
  String toString()
  String toString(int i, int radix)
  int compareTo(Integer anotherInteger)
  static int parseInt(String s)

class Double implements Comparable<Double>
  Double(double value)
  double doubleValue()
  boolean equals(Object anotherObject)
  String toString()
  int compareTo(Double anotherDouble)
  static double parseDouble(String s)

class String implements Comparable<String>
  int compareTo(String anotherString)
  boolean equals(Object anotherObject)
  int length()
  String substring(int begin)
    Returns substring(begin, length()).
  String substring(int begin, int end)
    Returns the substring from index begin through index (end – 1).
  int indexOf(String str)
    Returns the index within this string of the first occurrence of str. Returns
    –1 if str is not found.
  int indexOf(String str, int fromIndex)
    Returns the index within this string of the first occurrence of str, starting
    the search at fromIndex. Returns –1 if str is not found.
  int indexOf(int ch)
  int indexOf(int ch, int fromIndex)
  char charAt(int index)
  String toLowerCase()
  String toUpperCase()
  String[] split(String regex)
  boolean matches(String regex)
  String replaceAll(String regex, String str)

class Character
  static boolean isDigit(char ch)
  static boolean isLetter(char ch)
  static boolean isLetterOrDigit(char ch)
  static boolean isLowerCase(char ch)
  static boolean isUpperCase(char ch)
  static char toUpperCase(char ch)
  static char toLowerCase(char ch)

class Math
  static int abs(int a)
  static double abs(double a)
  static double pow(double base, double exponent)
  static double sqrt(double a)
  static double ceil(double a)
  static double floor(double a)
  static double min(double a, double b)
  static double max(double a, double b)
  static int min(int a, int b)
  static int max(int a, int b)
  static long round(double a)
  static double random()
    Returns a double greater than or equal to 0.0 and less than 1.0.
```

package java.util

```
interface List<E>
class ArrayList<E> implements List<E>
  boolean add(E item)
  int size()
  Iterator<E> iterator()
  ListIterator<E> listIterator()
  E get(int index)
  E set(int index, E item)
  void add(int index, E item)
  E remove(int index)

class LinkedList<E> implements List<E>, Queue<E>
  void addFirst(E item)
  void addLast(E item)
  E getFirst()
  E getLast()
  E removeFirst()
  E removeLast()

class Stack<E>
  boolean isEmpty()
  E peek()
  E pop()
  E push(E item)

interface Queue<E>
class PriorityQueue<E>
  boolean add(E item)
  boolean isEmpty()
  E peek()
  E remove()

interface Set<E>
class HashSet<E> implements Set<E>
class TreeSet<E> implements Set<E>
  boolean add(E item)
  boolean contains(Object item)
  boolean remove(Object item)
  int size()
  Iterator<E> iterator()
  boolean addAll(Collection<? extends E> c)
  boolean removeAll(Collection<?> c)
  boolean retainAll(Collection<?> c)

interface Map<K,V>
class HashMap<K,V> implements Map<K,V>
class TreeMap<K,V> implements Map<K,V>
  Object put(K key, V value)
  V get(Object key)
  boolean containsKey(Object key)
  int size()
  Set<K> keySet()
  Set<Map.Entry<K, V>> entrySet()

interface Iterator<E>
  boolean hasNext()
  E next()
  void remove()

interface ListIterator<E> extends Iterator<E>
  void add(E item)
  void set(E item)

class Scanner
  Scanner(InputStream source)
  Scanner(String str)
  boolean hasNext()
  boolean hasNextInt()
  boolean hasNextDouble()
  String next()
  int nextInt()
  double nextDouble()
  String nextLine()
  Scanner useDelimiter(String regex)
```

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

Package java.util.function

```
Interface BiConsumer<T,U>
  void accept(T t, U u)

Interface BiFunction<T,U,R>
  R apply(T t, U u)

Interface BiPredicate<T,U>
  boolean test(T t, U u)

Interface Consumer<T>
  void accept(T t)

Interface Function<T,R>
  R apply(T t)

Interface Predicate<T>
  boolean test(T t)

Interface Supplier<T>
  T get()
```

UIL COMPUTER SCIENCE WRITTEN TEST – 2026 INVITATIONAL A

Note: Correct responses are based on **Java SE Development Kit 25 (JDK 25)** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java SE 25 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using: `import static java.lang.System.*;`**

Question 1

Which of the following is the base 2 representation of 816_{10}?

A) 1100110000_2 **B)** 1101101101_2 **C)** 1101101000_2 **D)** 1100110001_2 **E)** 1101101010_2

Question 2

What is output by the code to the right?

A) 4 **B)** 5 **C)** 10 **D)** 11 **E)** 12

```
int result = 2 + 3 * 4 - 6 / 3;
out.println(result);
```

Question 3

What is output by the code to the right?

A) x = 5, y = 2.8 **B)** x = 5, y = 2.7

C) x = 5, y = 2.75 **D)** x = %d, y = %.1f

E) There is no output due to an error.

```
out.printf
    ("x = %d, y = %.1f\n", 5, 2.75);
```

Question 4

What is output by the code to the right?

A) Java **B)** Roc

C) Rocks **D)** Rock

E) ava

```
String s = "Java Rocks";
int len = s.length();
String t = s.substring(5, len);
out.println(t);
```

Question 5

What is output by the code to the right?

A) false **B)** true

```
boolean result = true || false && !false;
out.println(result);
```

Question 6

What is output by the code to the right?

A) -3.7 **B)** 3.7

C) 2.0 **D)** 2

```
double a = -3.7;
double b = 2.0;
double c = Math.abs(a);
out.println(Math.max(b, c));
```

Question 7

What is output by the code to the right?

A) 10 **B)** 11

C) 12 **D)** 9

E) 13

```
int x = 4;
x = x + 2 * 3;
x = x - 1;
out.println(x);
```

Question 8

What is output by the code to the right?

A) D

B) C

C) A

D) B

E) ABC

```
int score = 85;
if(score >= 90) {
    out.println("A");
} else if(score >= 80) {
    out.println("B");
} else {
    out.println("C");
}
```

Question 9

What is output by the code to the right?

A) `0 1 2 3`

B) `1 2 3 4`

C) `0 1 2 3 4`

D) `0 1 2`

E) `1 2 3`

```
for(int i = 1; i <= 3; i++) {
    out.print(i + " ");
}
```

Question 10

Which of the following correctly creates an `int` array of length 4 and sets its first element to `10`?

A)
```
int[] a = new int[4];
a[1] = 10;
```

B)
```
int[] a = new int[4];
a[0] = 10;
```

C)
```
int[4] a = {10};
```

D)
```
int a = new int[4];
a[0] = 10;
```

E)
```
int[] a = {10};
```

Question 11

Which standard `Scanner` method could not read the value `129` as input from a file?

A) `nextLine()` **B)** `next()` **C)** `nextInt()` **D)** `nextDouble()` **E)** `nextByte()`

Question 12

What is output by the code to the right?

A) `10`

B) `11`

C) `15`

D) `6`

E) `9`

```
int sum = 0;
for(int i = 1; i <= 4; i++) {
    sum += i;
}
out.println(sum);
```

Question 13

What is output by the code to the right?

A) `0` **B)** `2` **C)** `3` **D)** `1` **E)** `11`

```
int result = 3 + 4 * 2 > 10 ? 1 : 0;
out.println(result);
```

Question 14

Which of the following primitive data types can store the largest whole-number value?

A) `byte` **B)** `short` **C)** `int` **D)** `long` **E)** `char`

Question 15

Which of the following correctly declares and creates an `ArrayList` that stores `String` objects?

A) `ArrayList list = new ArrayList<String>();`

B) `ArrayList<String> list = new ArrayList<String>();`

C) `ArrayList<String> list = new ArrayList<string>();`

D) `ArrayList<String> list = new ArrayList<int>();`

E) None of the above.

Question 16

What is output by the code to the right?

A) `23`

B) `13`

C) `123`

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

```
try {
    int i = 1;
    i = i / --i;
    out.print(1);
} catch (ArithmeticException e) {
    out.print(2);
} finally {
    out.print(3);
}
```

Question 17

What could replace **<1*>** in the code to the right so that it will compile and run without error.

A) `push` **B)** `offer` **C)** `add`

D) Both B and C. **E)** All of the above.

Question 18

Assuming **<1*>** is filled in correctly, which of the following could not be found in the output by the code to the right?

A) `191` **B)** `15` **C)** `190` **D)** `13`

E) More than one of the above.

Question 19

What is the big $\mathcal{O}$ runtime of the code to the right

A) $\mathcal{O}(N \log \log N)$ **B)** $\mathcal{O}(N)$

C) $\mathcal{O}(N^2)$ **D)** $\mathcal{O}(\log \log N)$

E) $\mathcal{O}(N^2 \log \log N)$

```
Queue<Integer> s;
s = new LinkedList<Integer>();
for(int y = 0; y < 100; y++) {
    int i = 13;
    i += (int)(Math.random() * 178);
    s.<1*>(i);
}
while(!s.isEmpty())
    out.println(s.poll());
```

Question 20

Which of the lines in the code to the right first causes an error (compile- or run-time)?

A) `//1` **B)** `//2`

C) `//4` **D)** `//3`

E) None of the above. All lines will be error-free.

```
int i = 1, j = 0;
j *= 8.9; //1
i /= j++; //2
i += ++i; //3
j += (i -= 1); //4
```

Question 21

What is output if the following call is made to the function `fun` to the right? `fun(-7);`

A) `-7` **B)** `-212`

C) `0` **D)** `212`

E) There is no output due to a runtime error.

Question 22

What is output if the following call is made to the function `fun` to the right? `fun(14);`

A) `-83` **B)** `-89`

C) `-86` **D)** `-80`

E) There is no output due to a runtime error.

```
int fun(int n) {
    int i = 7;
    if(n < 0)
        return -212;
    if(n%2 == 1)
        i = 10;
    return i + fun(n - 1);
}
```

Question 23

What can replace **<1*>** in the code to the right so that the code compiles and runs without error?

A) `private` **B)** `abstract`

C) `parent` **D)** `public`

E) More than one of the above.

Question 24

What can replace **<2*>** in the code to the right so that the code compiles and runs without error?

A) Nothing is required. **B)** `super(m)`

C) `super()` **D)** A or C

E) Any of the above.

Question 25

What can replace **<3*>** in the code to the right so that the code compiles and the `getNum` method returns the sum of values `n` and `m`?

A) `m + super.getNum()`

B) `n + m`

C) `getNum() + super.getNum()`

D) A or B

E) All of the above.

Question 26

Assuming **<1*>**, **<2*>**, and **<3*>** are filled in correctly, what is output by the line marked `//q26` in the code to the right?

A) `17 17` **B)** `7 10`

C) `7 17` **D)** `17 10`

E) There is no output due to a runtime error.

Question 27

Assuming **<1*>**, **<2*>**, and **<3*>** are filled in correctly, what is output by the line marked `//q27` in the code to the right?

A) `null`

B) `null null`

C) The output is an empty line with one space.

D) There is no output due to a compile error.

E) There is no output due to a runtime error.

```
abstract class A {
    int n;
    String s;

    int getNum() {
        return n;
    }

    <1*> String getS();
}

class B extends A {
    int m;

    public B(int m) {
        <2*>;
        this.m = m;
    }

    String getS() {
        return s;
    }

    int getNum() {
        return <3*>;
    }
}

////////////// client code ////////////
A a = new B(7);
B b = new B(10);
String s = "" + a.getNum();
s += " " + b.getNum();
out.println(s); //q26
s = a.getS() + " ";
s += b.getS();
out.println(s); //q27
```

Question 28

What is output by the code to the right?

A) `false false` **B)** `false true`

C) `true false` **D)** `true true`

E) There is no output due to a compile error.

```
String mat = "theupsidedown";
String r = "[c-w]+\\D?";
String o = "" + mat.matches(r);
r = "\\w?{8,15}";
o += " " + mat.matches(r);
out.println(o);
```

Question 29

What is output by the code to the right?

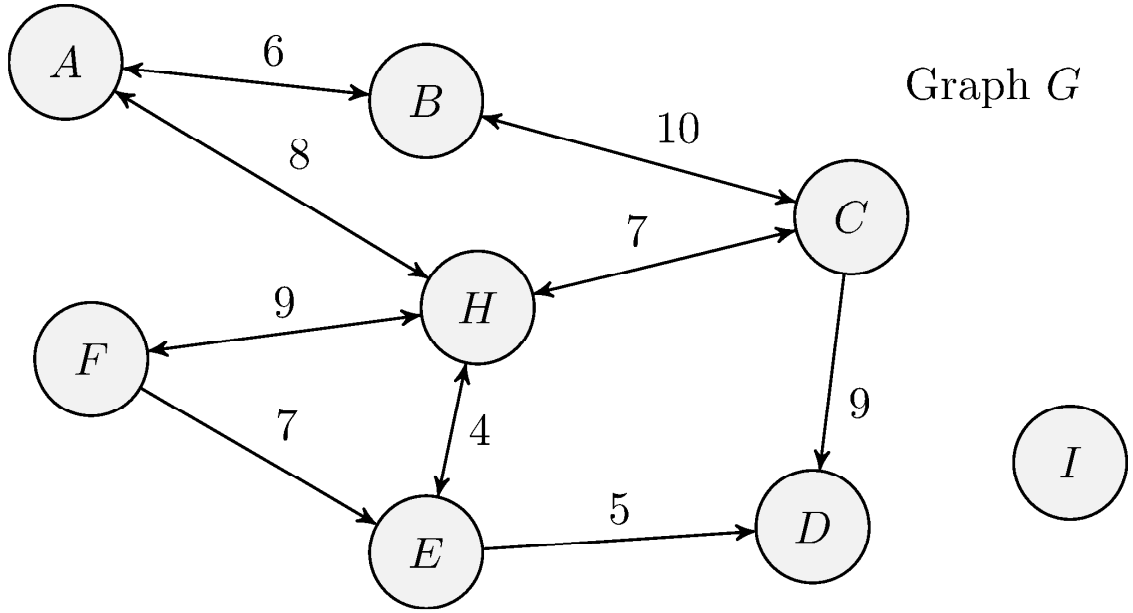
A) 98 **B)** 104

C) 106 **D)** 127

E) There is no output due to a runtime error.

```
int i = 0b1000101;
String s = Integer.toString(i, 6);
i = Integer.parseInt(s, 7);
s = Integer.toString(i, 9);
i = Integer.parseInt(s);
out.println(i);
```

Use the following graph for questions 30-33:



Question 30

Which of the following accurately describes graph G pictured above?

A) Undirected Unweighted **B)** Undirected Weighted **C)** Directed Weighted **D)** Directed Unweighted

Question 31

What is the cost of the shortest path from Node F to node B in graph G pictured above?

A) 23 **B)** 31 **C)** 25 **D)** 26 **E)** 28

Question 32

Which of the following describes Node I in graph G pictured above?

A) Isolated **B)** Sink **C)** Source **D)** All of the above. **E)** None of the above.

Question 33

What is the shortest cost path from Node A to node D in graph G pictured above?

A) $AHED$ **B)** $AHCD$ **C)** $AHFED$ **D)** $ABCD$ **E)** $ABCHED$

Question 34

What is output by the code to the right?

A) 8 **B)** 4

C) 2 **D)** 1

E) There is no output due to an error.

```
import static java.lang.System.*;

public class Q34 {
    public static final int NUM_BYTES = NUM_BITS / 8;
    public static final int NUM_BITS = Integer.SIZE;
    public static void main(String[] args) {
        out.println(NUM_BYTES);
    }
}
```

Question 35

What is output by the code to the right?

A) `8` **B)** `4` **C)** `2`

D) There is no output due to a compilation error on `L1`.

E) There is no output due to a compilation error on `L2`.

```
import static java.lang.System.*;

public class Q35 {
    public final int NUM_BYTES = NUM_BITS / 8; // L1
    public static final int NUM_BITS = Long.SIZE;
    public static void main(String[] args) {
        out.println(NUM_BYTES); // L2
    }
}
```

Question 36

What is output by the code to the right?

A)
```
5
1_000_000_009
```

B)
```
5
1000000009
```

C)
```
1_000_000_009
5
```

D)
```
1000000009
5
```

E) There is no output due to an error.

Question 37

Which of the following changes to the program to the right will cause the program to run without error?

A) Do nothing. The code will run without error as is.

B) Remove the underscores from L1.

C) Change `L2` to `System.out.println(5);`.

D) Wrap `L2` in a `static` block (i.e., `static { … }`).

E) Remove `L3`.

```
import static java.lang.System.*;

public class Q36_37_38 {
    private static final int MOD = 1_000_000_009; // L1

    out.println(5); // L2

    public static void main(String[] args) {
        out.println(MOD); // L3
    }
}
```

Question 38

Assuming that the correct choice from question 37 has been implemented in the code to the right, what will be the output of the modified program?

A)
```
5
1_000_000_009
```

B)
```
5
1000000009
```

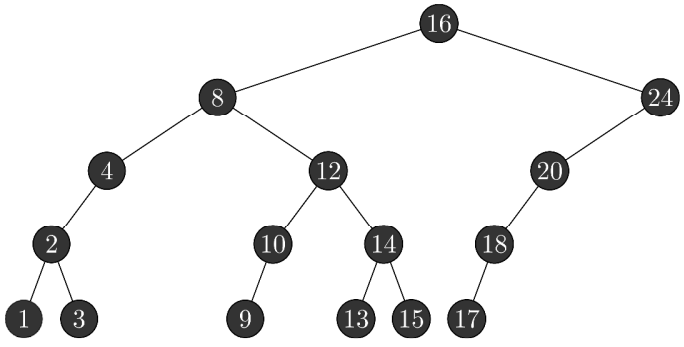
C)
```
1_000_000_009
5
```

D)
```
1000000009
5
```

E) None of the above.

Question 39

What is the minimum number of nodes that need to be ***added*** to the tree to the right to make it a ***full*** binary tree? Write your answer in the blank provided on your answer sheet for this question.



Question 40

What is the minimum number of nodes that need to be ***added*** to the tree to the right to make it a ***complete*** binary tree? Write your answer in the blank provided on your answer sheet for this question.